Question			Answer	Marks	AOs		Guidance
12	(a)		$2p + q + 0.2 + 0.3 = 1$ **soi oe**	**B1**	**2.1**		
			$2 \times p \times q = 0.06$ **soi**	**M1**	**3.1a**	allow **M1** if 2 omitted	
			eliminate p or q with a correct substitution from one of *their* equations	**M1**	**1.1**		
			$q^2 - 0.5q + 0.06 = 0$ or $2p^2 - 0.5p + 0.03 = 0$ **oe**	**A1**	**1.1**	eg $2 \times \frac{0.03}{q} + q = 0.5$ or $2p + \frac{0.03}{p} = 0.5$	**NB** if 2 omitted, **A0** for $2p^2 - 0.5p + 0.6 = 0$ or $2q^2 + q + 0.24 = 0$ which have no real roots
			$q = 0.2$ or 0.3 **and** $p = 0.15$ or 0.1	**A1**	**1.1**	may be implied by eg $q = 0.2$ or 0.3 **and** $2p = 0.3$ or 0.2	
			($q < 2p$ so) $q = 0.2$ and $p = 0.15$	**A1**	**3.2a**		
				[6]			
12	(b)		$10 \times q \times (1 - q)^9$ **soi**	**M1**	**1.1**		
			0.27 or 0.268 or awrt 0.2684 **isw**	**A1**	**1.1**	**FT** *their* q where $0 < q < 1$	
				[2]			

Question			Answer	Marks	AOs		Guidance
12	(c)		H_0: $p = 0.2$ H_1: $p > 0.2$	**B1**	**1.1**	both hypotheses; allow equivalent in words or eg P(1) = 0.2	allow any parameter as long as clearly defined as probability
			p is the probability that the spinner shows a 1 (on any given spin) **oe**	**B1**	**2.5**		
			use of $X \sim B(100,0.2)$ where x is the number of 1s obtained in 100 spins to obtain $P(X \geq k)$ or $P(X \leq k)$	**M1**	**3.3**	k = 27, 28 or 29	**M0** for $P(X = k)$ **NB** $P(X = 28) = 0.014\ldots$ $P(X = 27) = 0.020168\ldots$
			$P(X \leq 27) =$ awrt 0.97 or $P(X \geq 28) =$ awrt 0.034	**A1**	**1.1**	or critical region is $X \geq 28$	
			0.034 < 0.05 or 0.97 > 0.95	**M1**	**3.4**	or 28 is in critical region	**FT** *their* probability, dependent on award of first **M1**
			significant or reject H_0 or accept H_1; may be embedded in conclusion in context	**A1**	**1.1**	must have the correct probability or correct critical region for the last two **A** marks	
			there is sufficient evidence to **suggest** (at 5% level) that the probability of a score of 1 is greater than 0.2	**A1**	**2.2b**		do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; **A0** if answer spoiled
				[7]			